SAHIL AHAMED SHAIK

Robotics enthusiast



- **EDUCATION:**
  - B.tech-mechatronics from Hindustan Institute of Technology and Sciences (HITS).

- **EXPERIENSES:**
  - Abhyaz internship (Robotics intern)
    - Learning about ROS, SolidWorks, and CAD software
    - Self-learning approach.
  - Edgeforce solutions (ROS intern)
  - Worked on UGV path-planning (MOVE_BASE PACKAGE + creating self program)
    - Worked on CBM designing and developing along with UGV parts
    - Worked on EXO skeleton.

- **PUBLICATIONS AND PRESENTATIONS:**
  - Design and evaluation of various knee assistive mechanisms for rehabilitation design and evaluation of various knee assistive mechanisms for rehabilitation, AIP Publishing · Nov 10, 2023
  - Presented at RIACT 2022, RANE 2023 and AIIMS- 3.0.

PROJECTS DONE

- **Designing and developing of autonomous sanitizing robot**
  - Learned the basic circuit and sensor details
  - The development of robot body also studied

- **Design and development of a homogeneous swarm of cooperative robots in a virtual environment**
  - The robot communication methods explored
  - The inner robot communication is established using the central node

- **Design and development of knee rehabilitation robot**
  - Designed more than five SolidWorks designs and developed three models for knee rehabilitation
  - Infused AI to detect knee osteosis
  - Patent in progress and publication is done

- **Design and Development of Borewell Rescue System**
  - Designed robotic arm in SolidWorks
  - 2 designs have been designed and one design was developed

MAJOR SKILLS

- ROS 1&2 – GAZEBO, RVIZ, MOVEIT
- SolidWorks, Fusion 360
- ANSYS
- Python, C++
- AI(tensor flow)
- Unity (AR/VR)
- MATLAB
- CAD software

CERTIFICATIONS

- Completed 3 days of SMC training in pneumatics.
- Completed TensorFlow 2.0: Deep Learning and Artificial Intelligence-Udemy.
- Completed ROS for Beginners: Basics, Motion, and OpenCV- Udemy.
- Completed Introduction to AI, Machine Learning, and Python basics- Udemy. Completed MATLAB and Simulink onramp- MathWorks training service
- In progress :
  - RPA
  - Reinforcement learning - courses

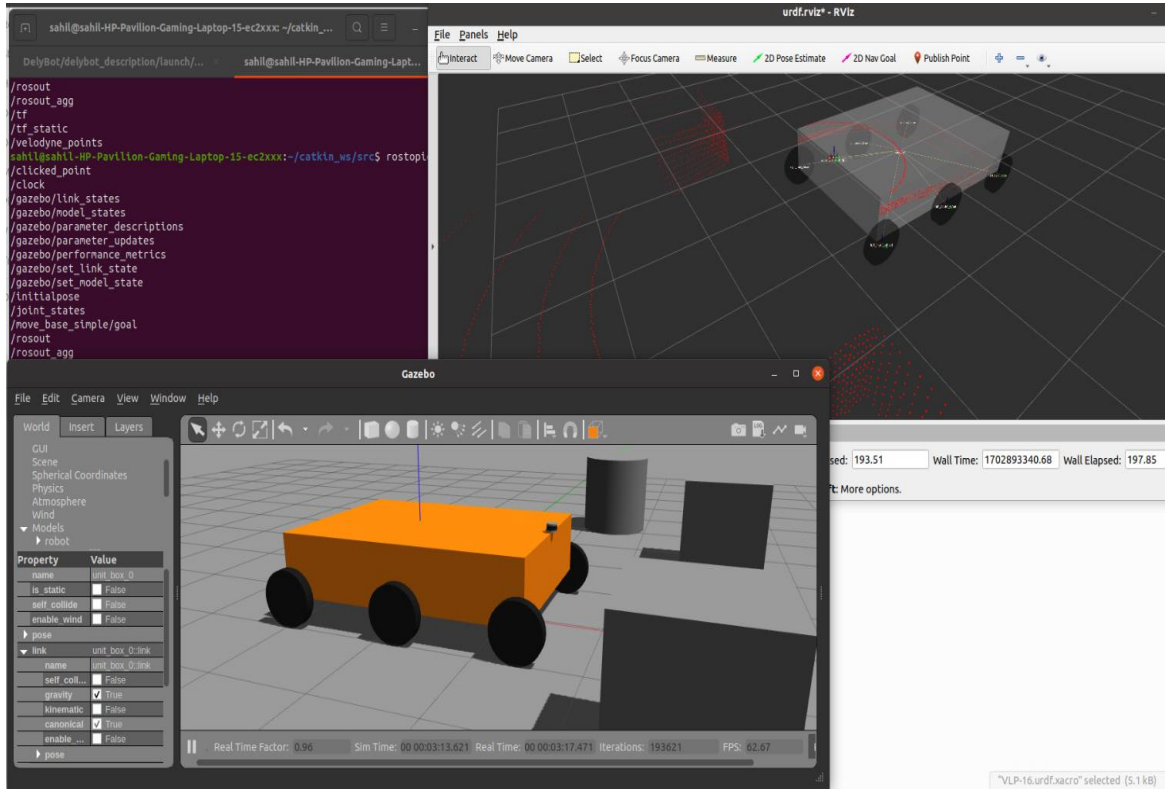
UGV Path-Planning

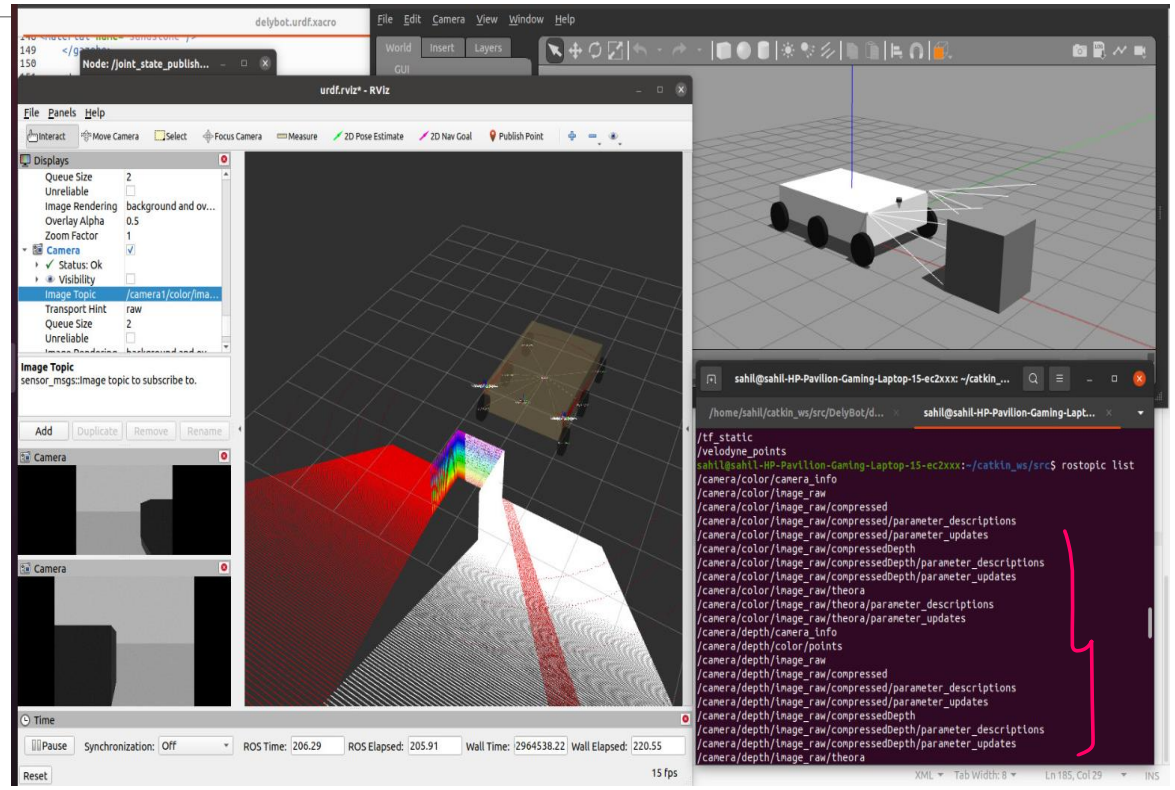
Edgeforce solutions(experience)

UGV Path-Planning

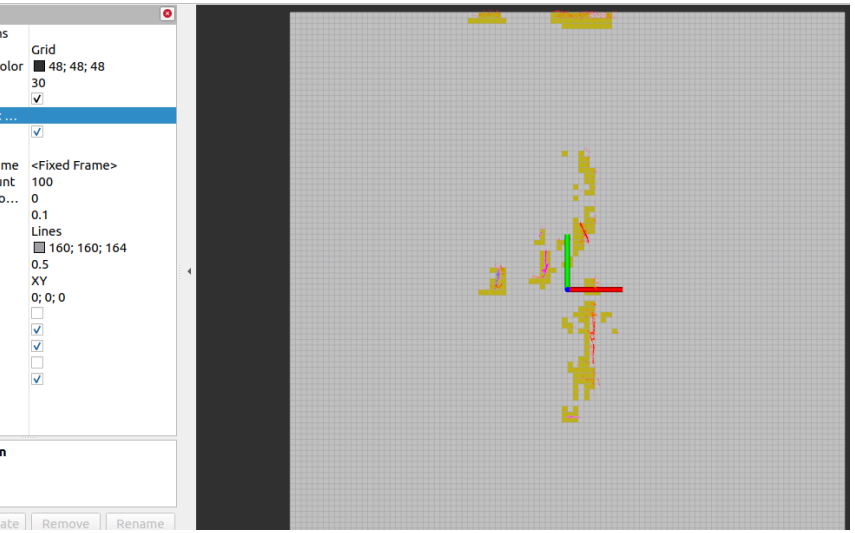
- This is the static programming of the robot where the robot movement is defined by giving waypoints at the start.
- The major obstacle detention is done by 3D point cloud sensor and the camera is there for region of interest.
- The two methods described are
  - Move Base package using DWA and trajectory planning
  - The base programming using A* along with local and global map
- The simulations are done in the gazebo and RVIZ for initial testing. The sensor data collection along with final testing was done in the UGV with some major changes.
- Along with that airport tug designing and PLANSYS2 in ROS 2.

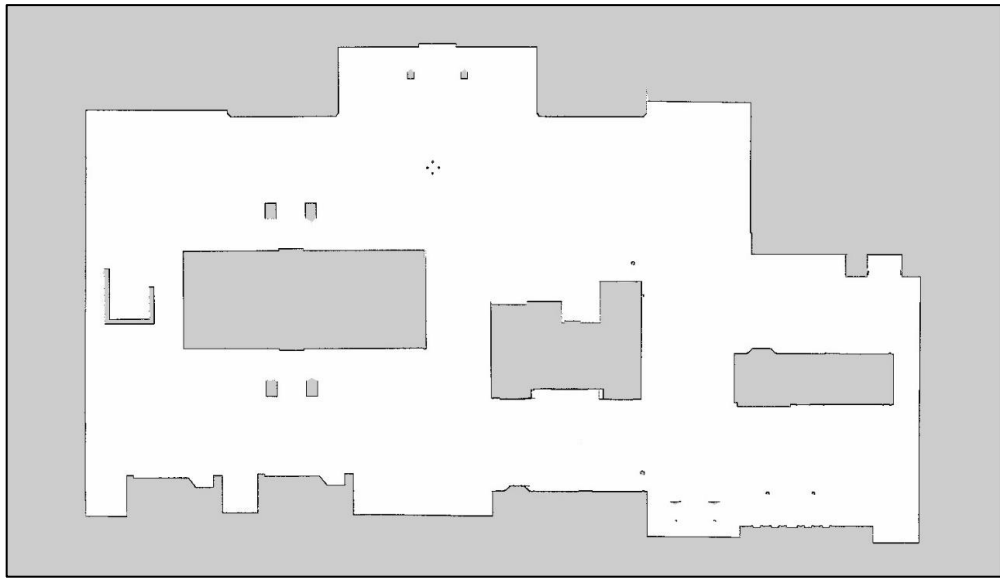
ROS INSTALLATION AND CAMERA INTEGRATION TO TUG IN RVIZ+GAZEBO

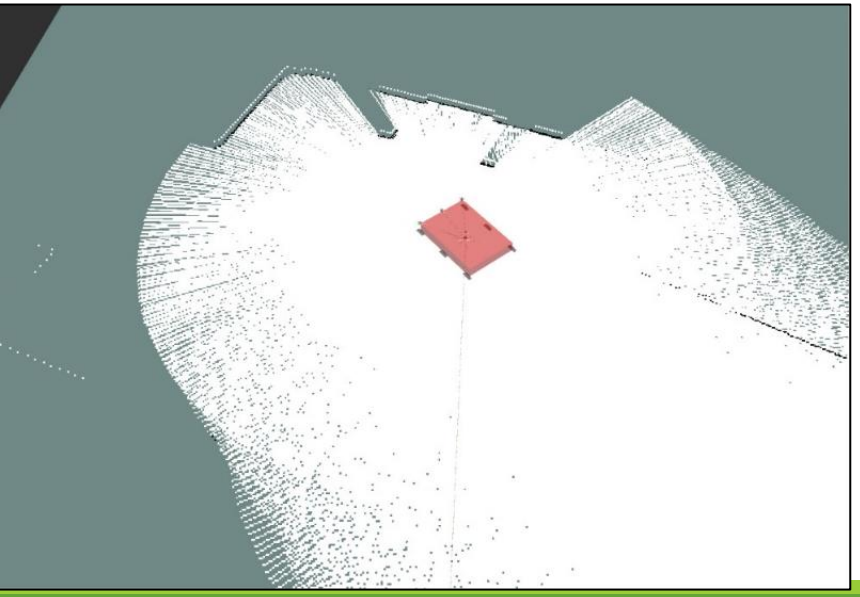


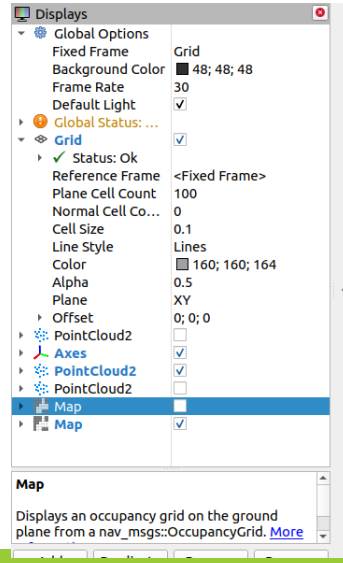


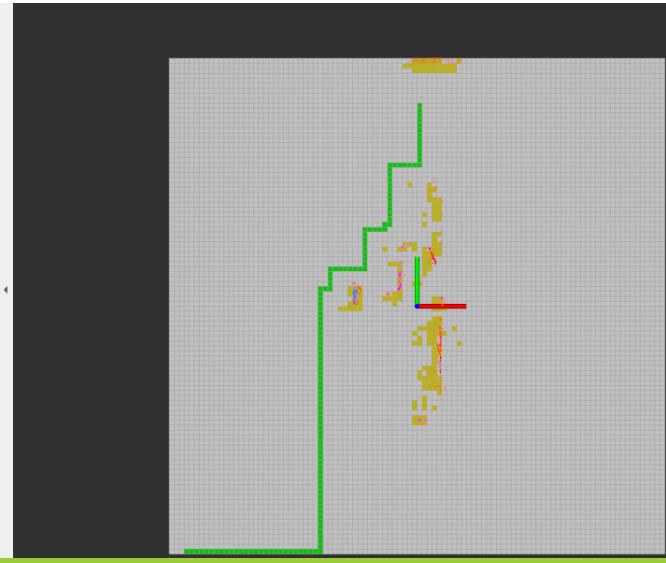
Path planning using A* and SLAM

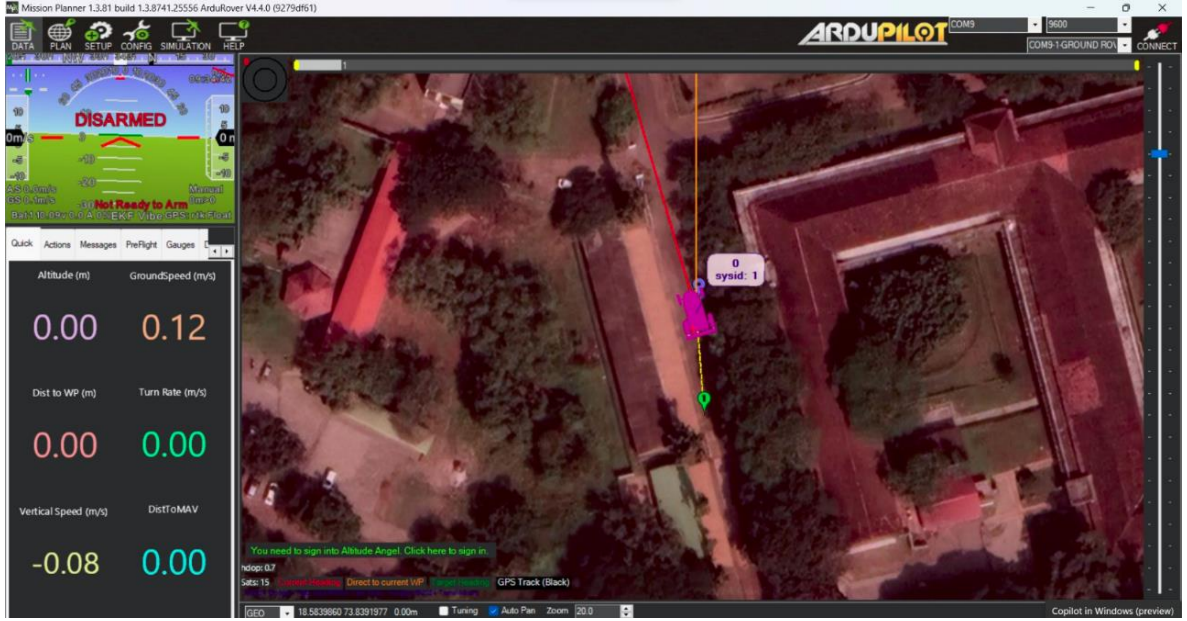


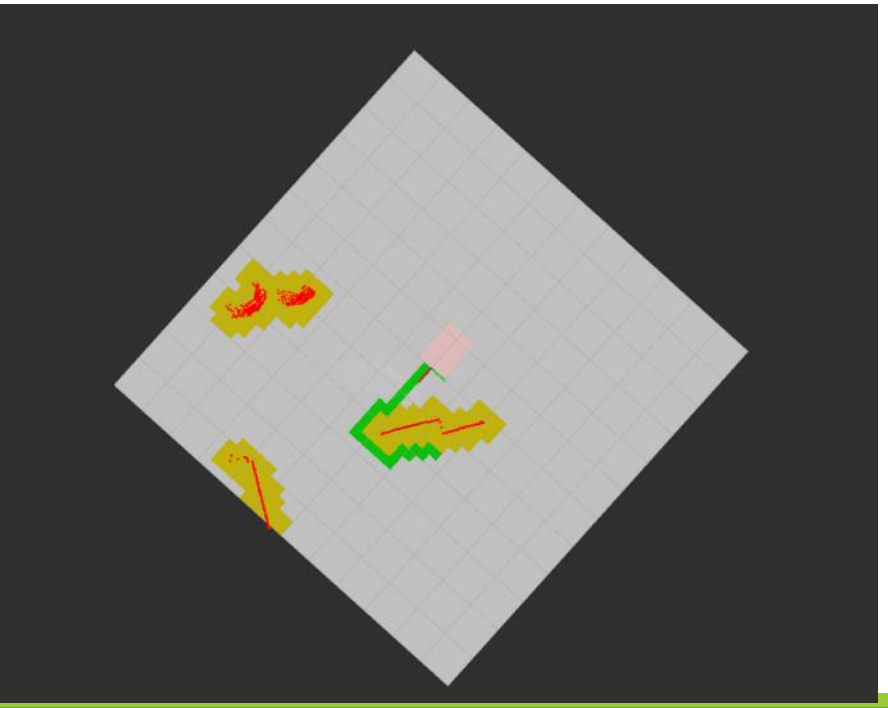




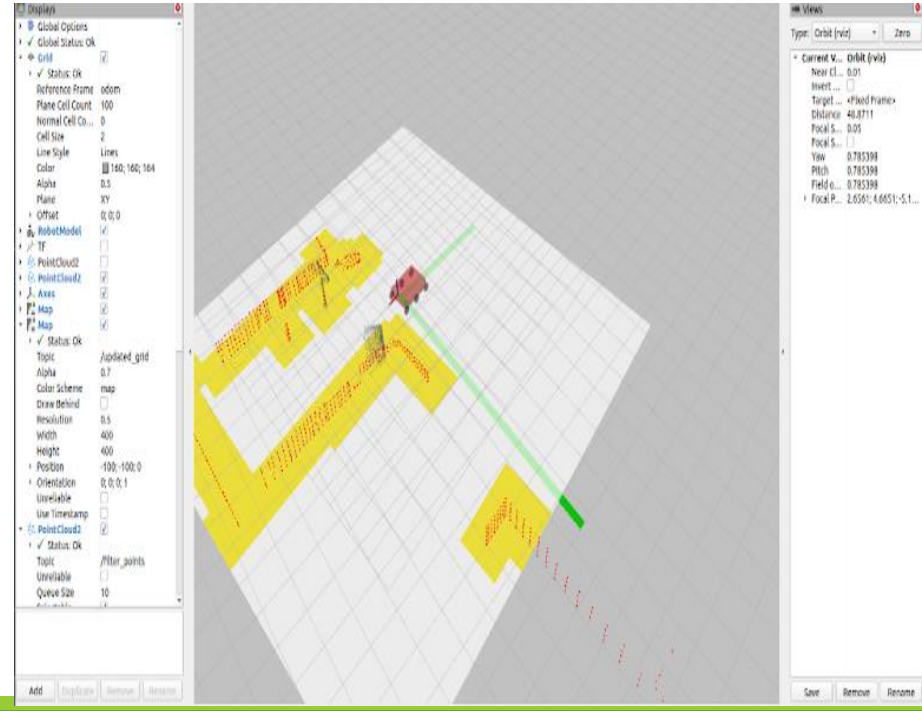




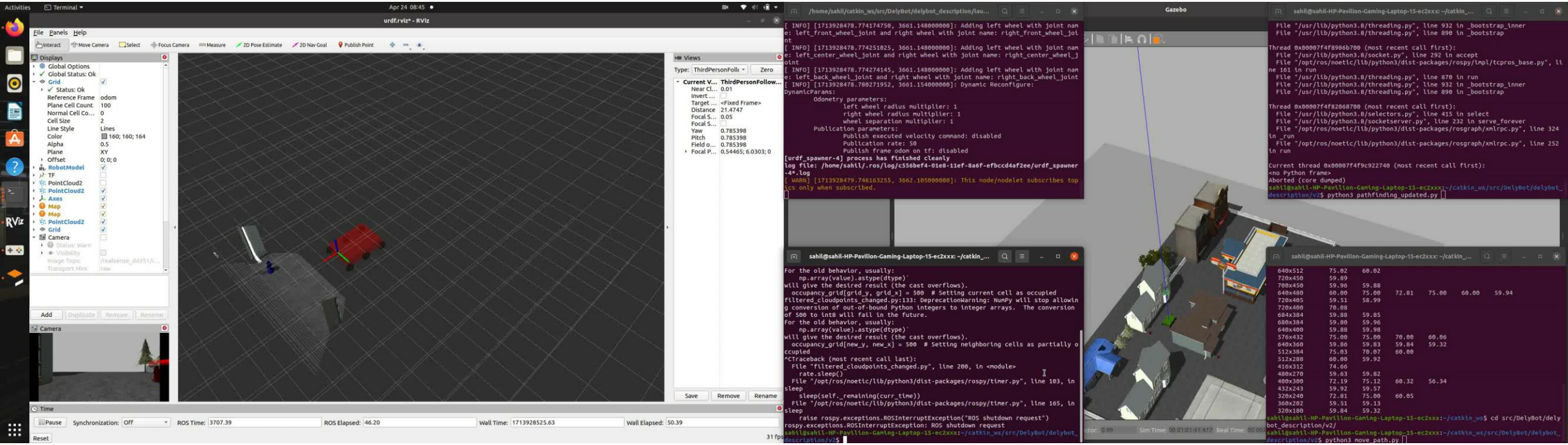


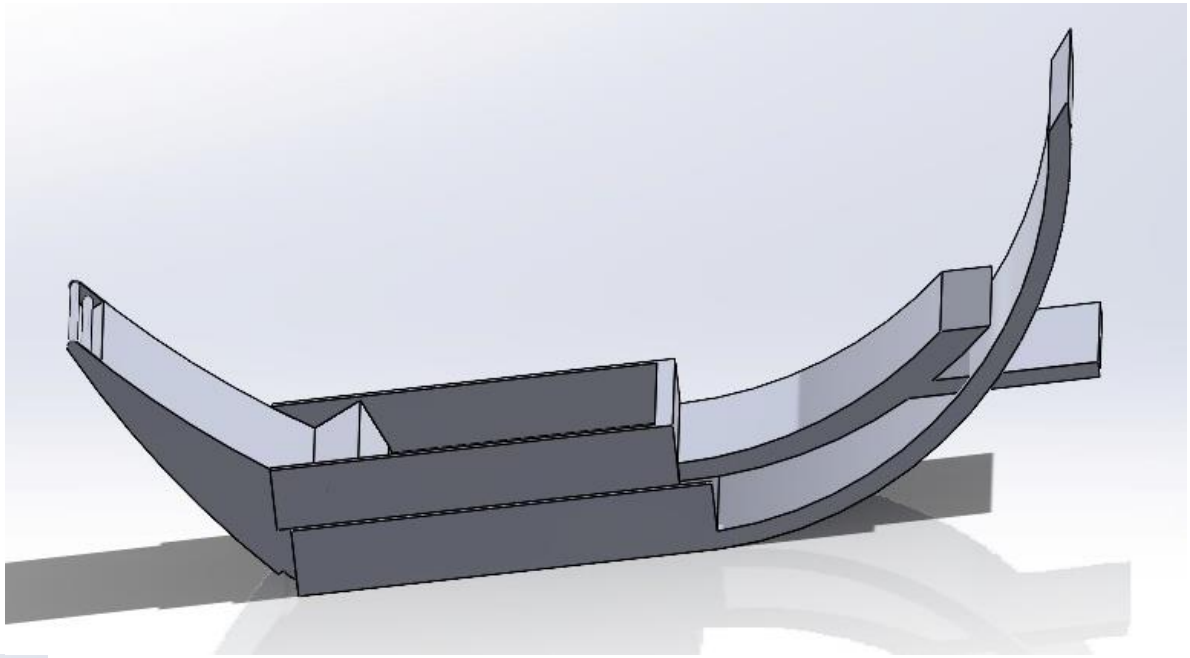


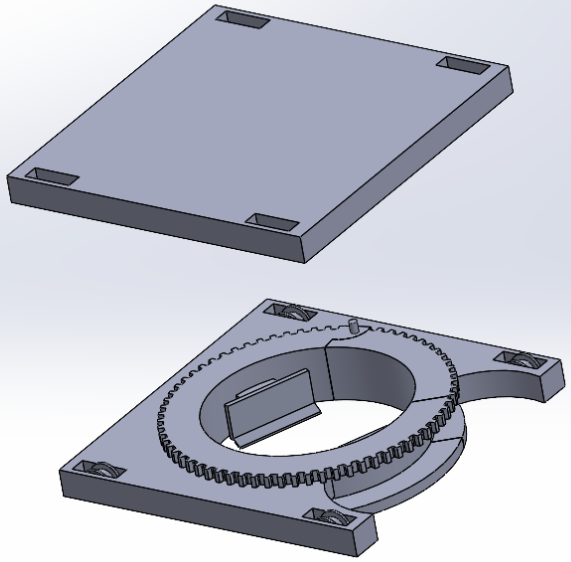
A* waypoint follower algorithm



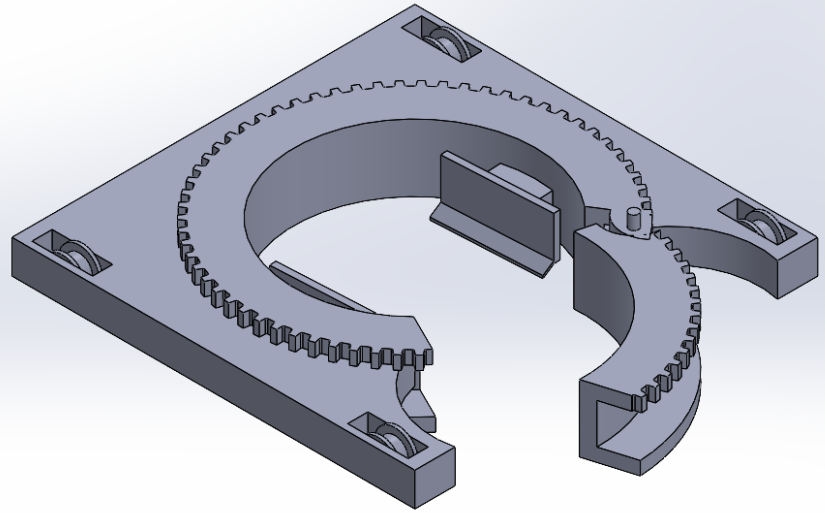
Path-planning algorithm







AIRPORT TUG DESIGNS

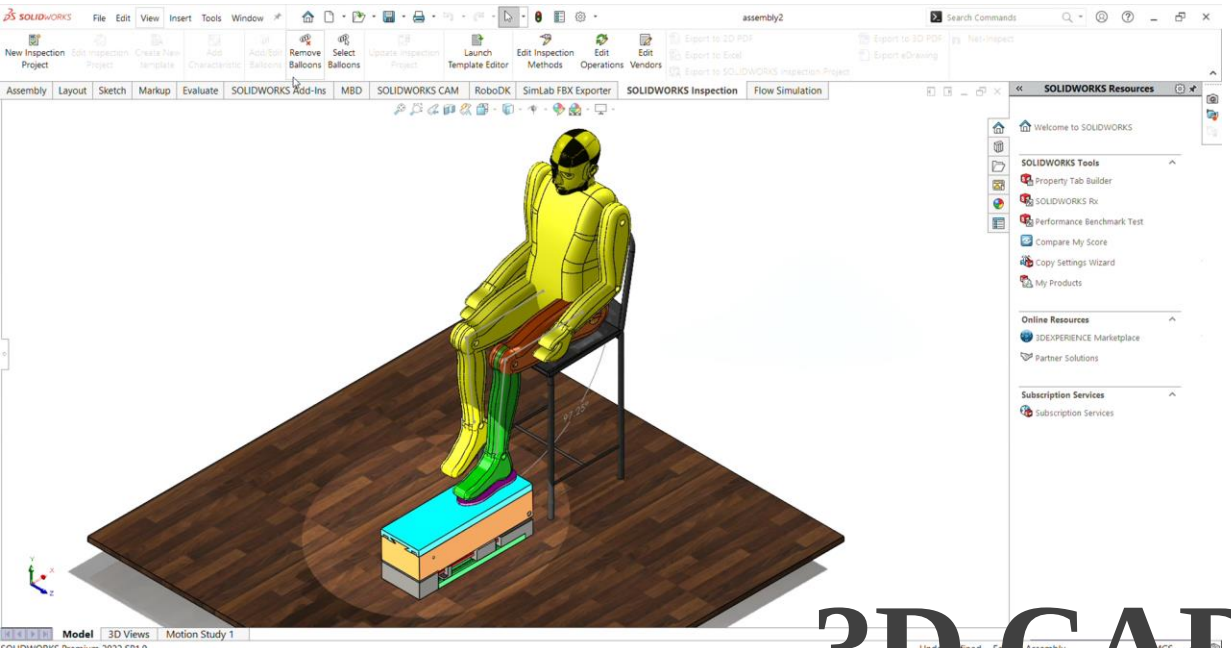


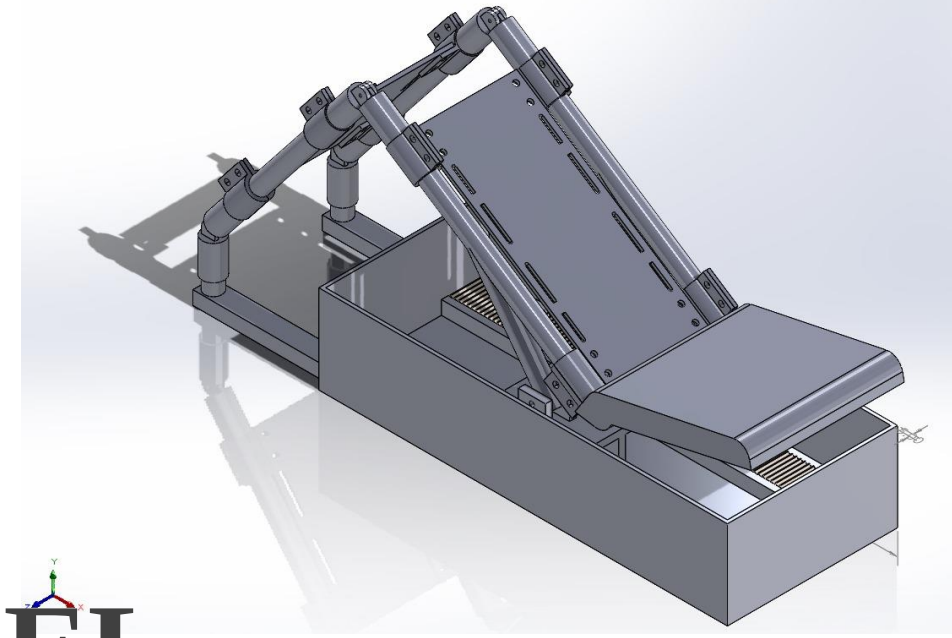
Design and development of knee rehabilitation robot

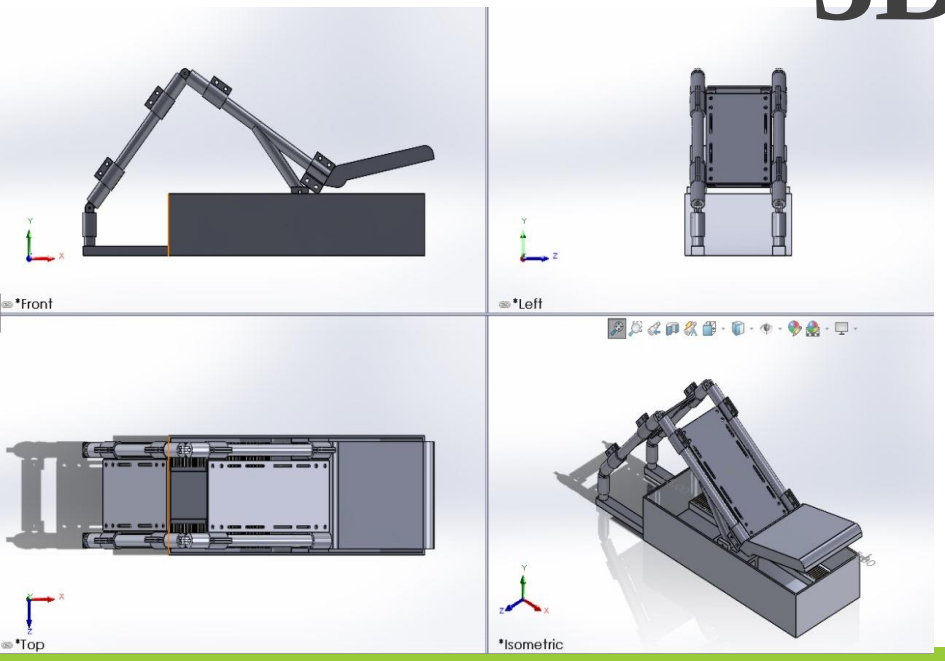
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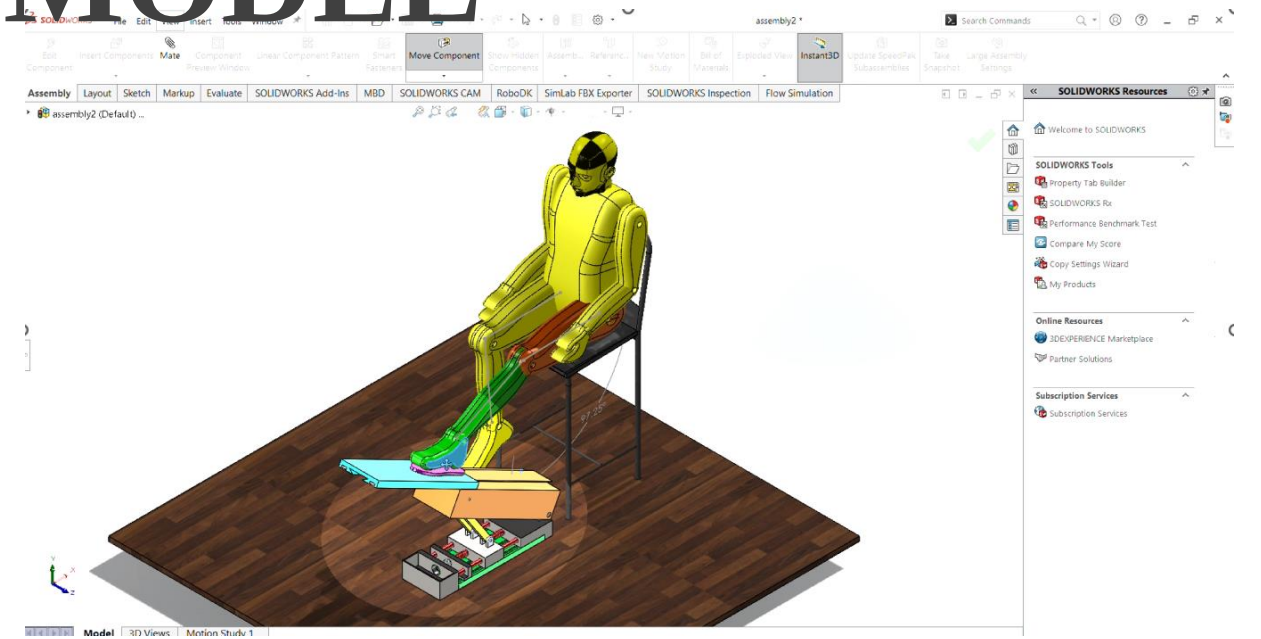
- The robot is designed to give physiotherapy in the absence of the doctors. The robot's mechanical design is to hold 2 degrees of motion and give an arc movement instead of an X-Y movement.
- The robot had to undergo extensive research for available designs and their advantages & disadvantages. To find the best optimal design to mimic human action.
- AI algorithm was applied to check the patients had knee osteosis, using the convolution algorithm.
- One design is going through a patent and the other is going to be published and hopefully be patent in the coming future

3D CAD MODEL

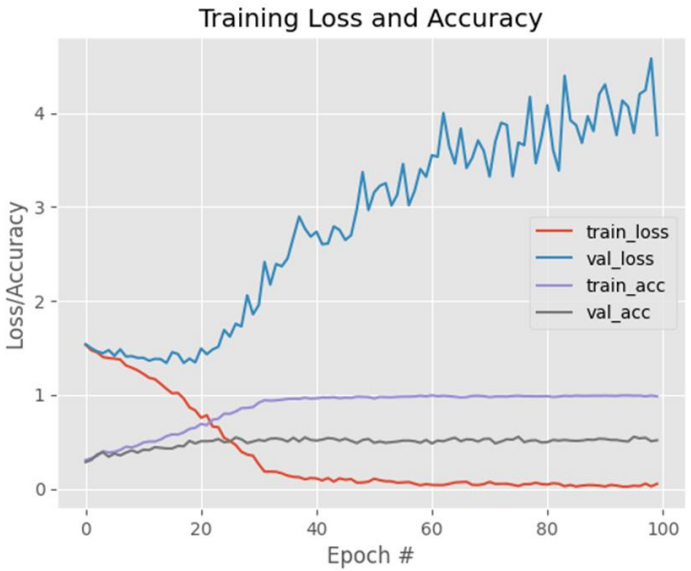

SOLIDWORKS Resources
Welcome to SOLIDWORKS
SOLIDWORKS Tools
Property Tab Builder
SOLIDWORKS Rx
Performance Benchmark Test
Compare My Score
Copy Settings Wizard
My Products
Online Resources
3DEXPERIENCE Marketplace
Partner Solutions
Subscription Services
Subscription Services
Model
3D Views
Motion Study 1
SOLIDWORKS Premium 2022 SP1.0

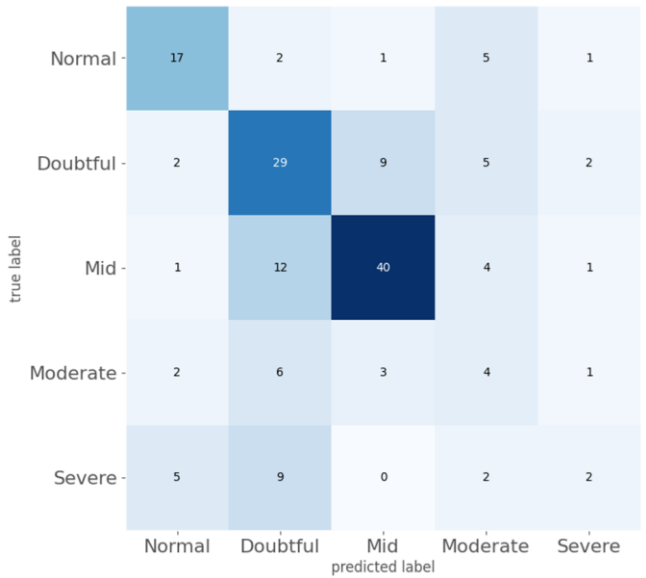



*Front
*Left
*Top
*Isometric


Assembly
Layout
Sketch
Markup
Evaluate
SOLIDWORKS Add-Ins
MBD
SOLIDWORKS CAM
RoboDK
SimLab FBX Exporter
SOLIDWORKS Inspection
Flow Simulation
SOLIDWORKS Resources
Welcome to SOLIDWORKS
SOLIDWORKS Tools
Property Tab Builder
SOLIDWORKS Rx
Performance Benchmark Test
Compare My Score
Copy Settings Wizard
My Products
Online Resources
3DEXPERIENCE Marketplace
Partner Solutions
Subscription Services
Subscription Services
Model
3D Views
Motion Study 1

AI ALGORITHM OUTPUT(Relu activation)





Activation Function	Epoch	Test loss	Test accuracy
Relu	100	3.48	0.56
Relu	50	2.82	0.53
Linear	100	3.71	0.50
Linear	50	3.11	0.50
LeakyRelu	100	4.58	0.52
LeakyRelu	50	3.08	0.52

`test loss: 3.478864908218384 %`

`test accuracy: 0.5575757622718811 %`

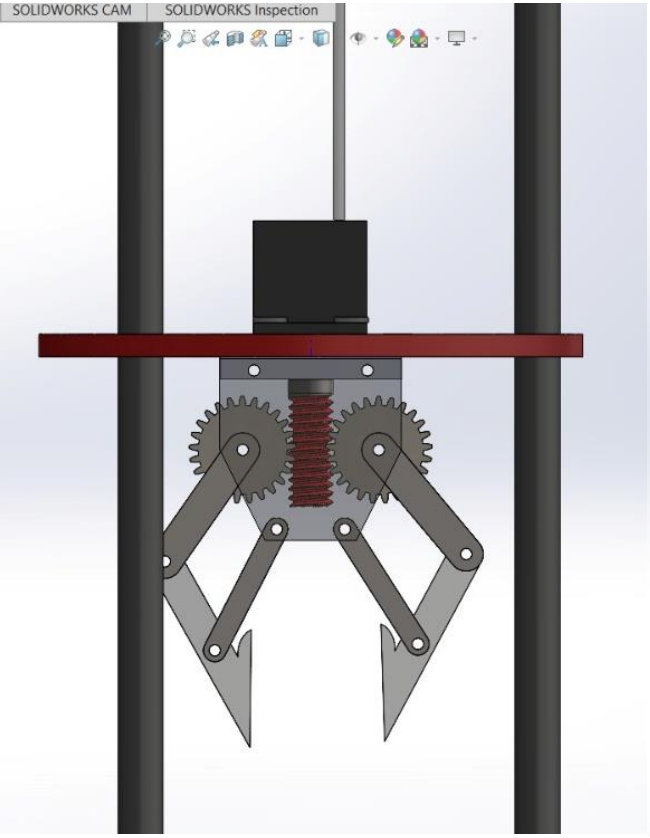
Source Code Link - https://colab.research.google.com/drive/1uN9005P1ps0T7iTNDYV62JI_ArKktOZW#scrollTo=XoKGbeNYJXFr

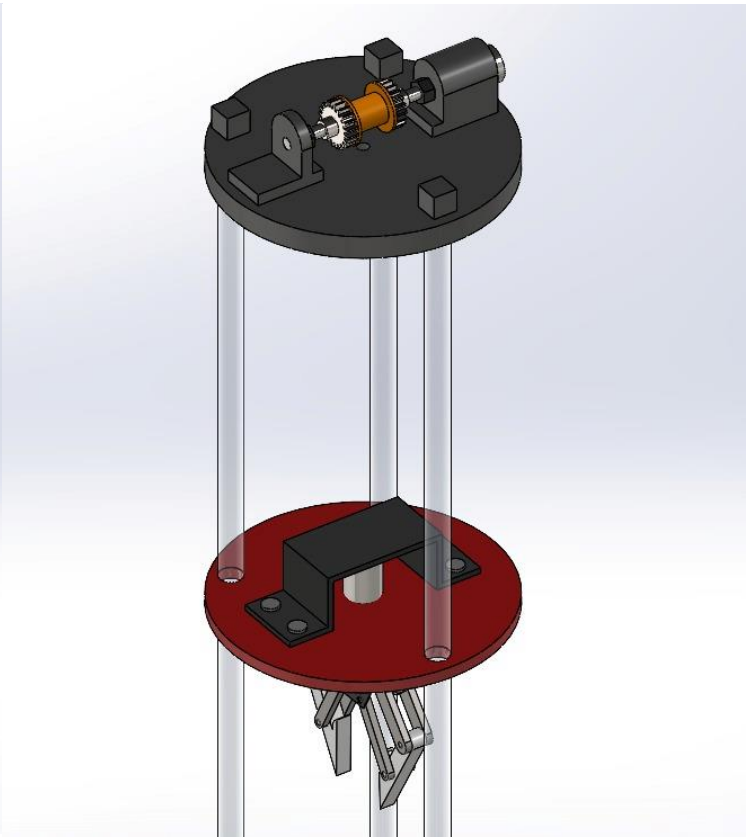
Design and Development of Borewell Rescue System

Design and Development of Borewell Rescue System

- The borewell system was designed to save children who fall into the borewell. The robot has soft grippers to rescue them without harming them
- The design was done keeping the dimensions accuracy and the joint functionality.
- The designing major motto was to keep the design simple and affordable.
- The major designs were taken to be as market-available and programable content was also made to design and upload within minutes so as of not to waste time.
- The final and initial designs as in the below slide along with simulation.

BOREWELL RESCUE ROBOT







Initial design



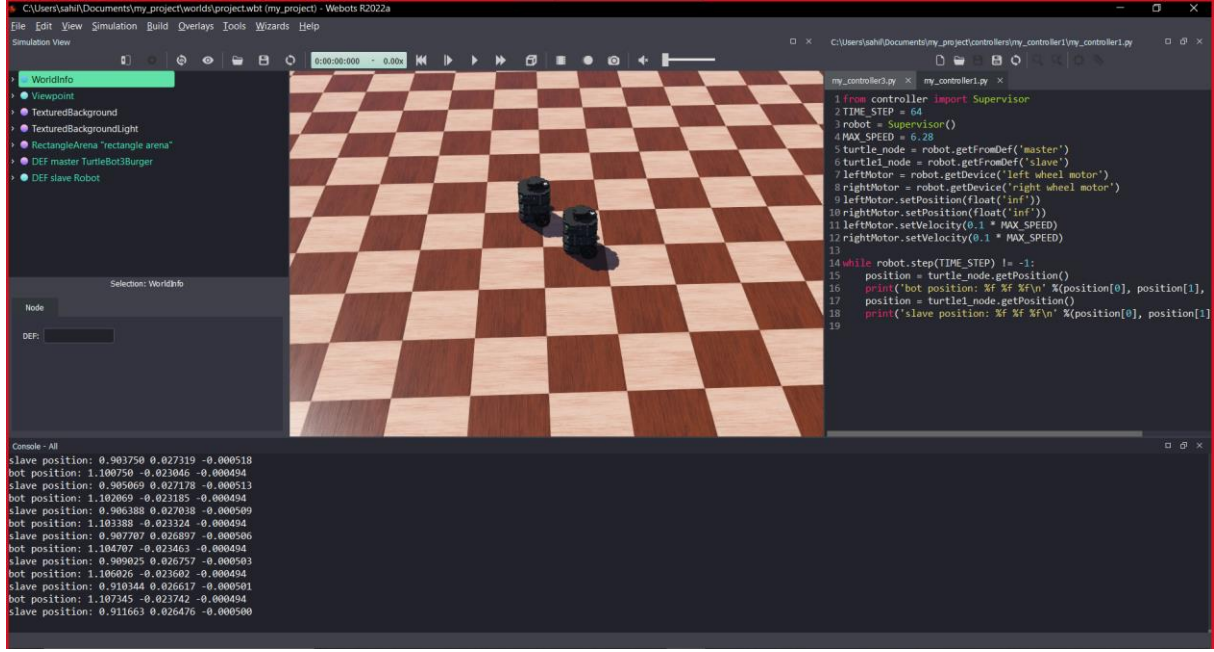
SIMULATION OF ROBOT

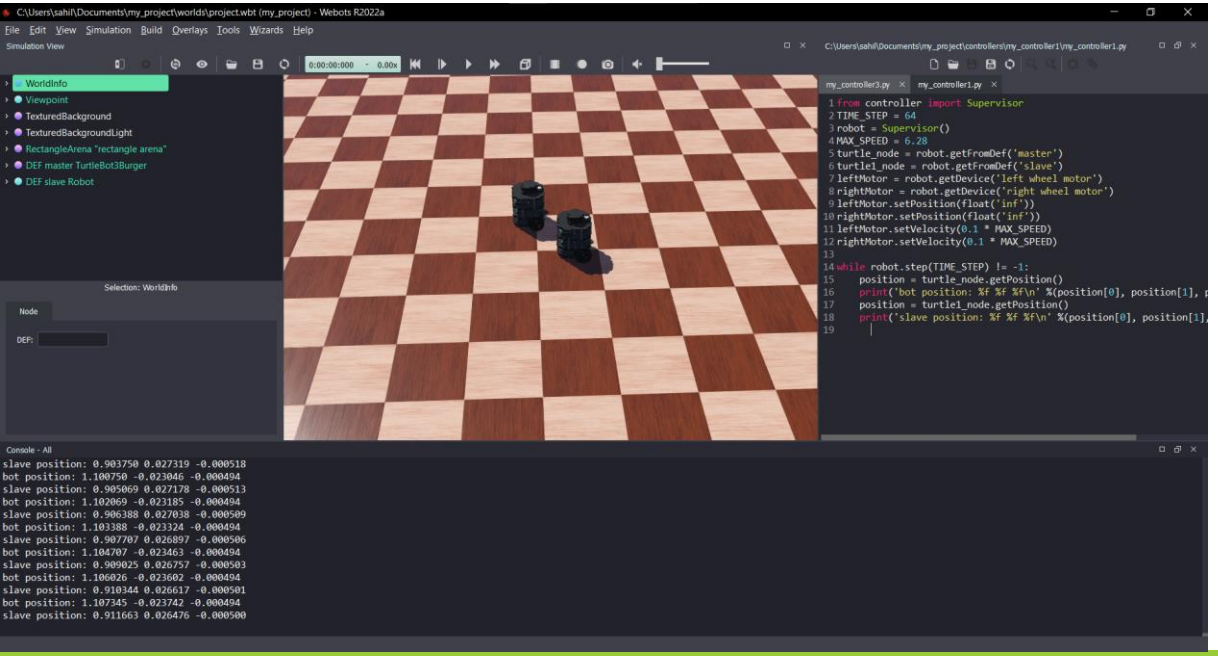


Design and development of a homogeneous swarm of cooperative robots in a virtual environment

Design and development of a homogeneous swarm of cooperative robots in a virtual environment

- The homogeneous robot communications were explored and studied in detail the algorithms and formulas related to the swarm robotics.
- The Webot software provides a similar experience as of ROS along with prebuilt packages to learn to code in whichever language is comfortable.
- The robots are communicating with each other using the world node as the medium the speed is getting communicating with each other hence movement is possible
- The robot adding the cumulative movement in the X-Y plane is done and the output was not satisfactory as it needed interrupt control to avoid obstacles first.

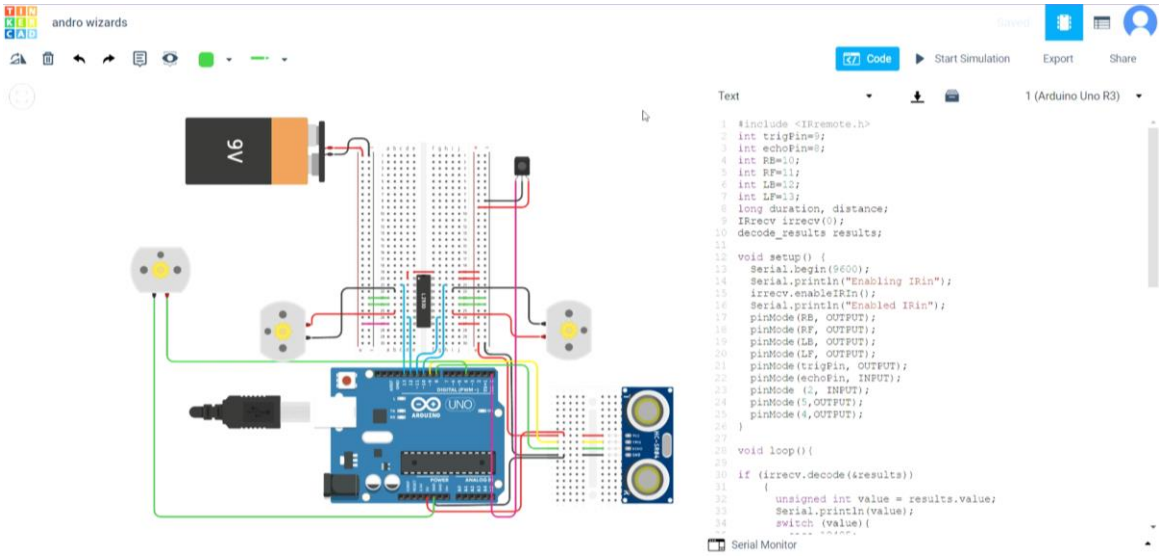

C:\Users\sahil\Documents\my_project\worlds\project.wbt (my_project) - Webots R2022a
File Edit View Simulation Build Overlays Tools Wizards Help
Simulation View
WorldInfo
Viewpoint
TexturedBackground
TexturedBackgroundLight
RectangleArena "rectangle arena"
DEF master TurtleBot3Burger
DEF slave Robot
Selection: WorldInfo
Node
DEF:
my_controller3.py
my_controller1.py
from controller import Supervisor
TIME_STEP = 64
robot = Supervisor()
MAX_SPEED = 6.28
turtle_node = robot.getFromDef('master')
turtle1_node = robot.getFromDef('slave')
leftMotor = robot.getDevice('left wheel motor')
rightMotor = robot.getDevice('right wheel motor')
leftMotor.setPosition(float('inf'))
rightMotor.setPosition(float('inf'))
leftMotor.setVelocity(0.1 * MAX_SPEED)
rightMotor.setVelocity(0.1 * MAX_SPEED)
while robot.step(TIME_STEP) != -1:
position = turtle_node.getPosition()
print('bot position: %f %f %f\n' %(position[0], position[1],
position = turtle1_node.getPosition()
print('slave position: %f %f %f\n' %(position[0], position[1]
Console - All
slave position: 0.903750 0.027319 -0.000518
bot position: 1.100750 -0.023046 -0.000494
slave position: 0.905069 0.027178 -0.000513
bot position: 1.102069 -0.023185 -0.000494
slave position: 0.906388 0.027038 -0.000509
bot position: 1.103388 -0.023324 -0.000494
slave position: 0.907707 0.026897 -0.000506
bot position: 1.104707 -0.023463 -0.000494
slave position: 0.909025 0.026757 -0.000503
bot position: 1.106026 -0.023602 -0.000494
slave position: 0.910344 0.026617 -0.000501
bot position: 1.107345 -0.023742 -0.000494
slave position: 0.911663 0.026476 -0.000500

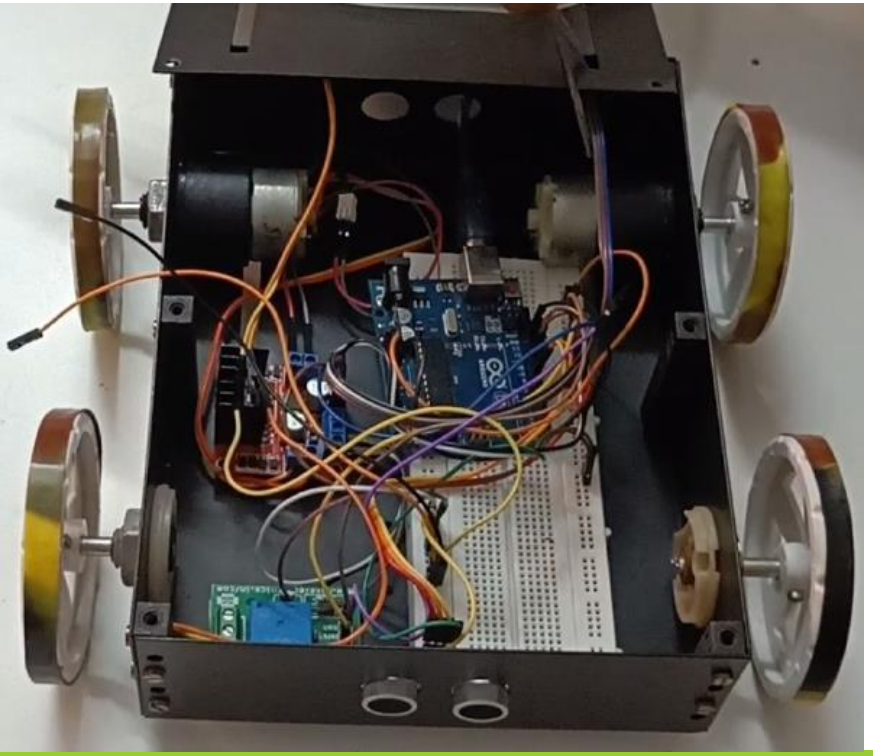

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bot position: 1.107345 -0.023742 -0.000494
slave position: 0.911663 0.026476 -0.000500

Designing and developing of autonomous sanitizing robot

Designing and developing of autonomous sanitizing robot

- The robot was visioned during the lockdown time to give to the shopkeepers to sanitize their shops at regular intervals to ensure sanitized shops.
- The robot uses a basic avoidance program by using ultrasonic sensor data.
- The robot has an IR sensor which is limited by distance to detect objects for the spray to initialize and spray on the cupboard of the shops.
- The robot chassis was tested to hold the water tank and the program simulation was made in Tinkercad fro simulation and code testing.





Sanitization robot



THANK YOU



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https://github.com/sahil86469